12.23 Stirling from Probability

WORKSHEET

The question

(i) The random variable X has a binomial distribution with parameters n and p, where $n = 16$ and $p = \frac{1}{2}$. Show, using an approximation in terms of the standard normal density function $\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$, that

$$\mathrm{P}(X = 8) \approx \frac{1}{2\sqrt{2\pi}}.$$

(ii) By considering a binomial distribution with parameters $2n$ and $\frac{1}{2}$, show that

$$(2n)! \approx \frac{2^{2n}(n!)^2}{\sqrt{n\pi}}.$$

(iii) By considering a Poisson distribution with parameter n, show that

$$n! \approx \sqrt{2\pi n}\, e^{-n} n^n.$$

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If a discrete random variable X with mean μ and variance σ^2 is approximately normal, roughly what is $\mathrm{P}(X = \mu)$? (Think of the probability of an interval of width 1 centred at μ.)

STAGE 2 The central binomial coefficient

Ask yourself For $\mathrm{B}(2n, \frac{1}{2})$, what are the mean and standard deviation? Write $\mathrm{P}(X = n)$ exactly and approximately.

STAGE 3 Stirling's formula

Ask yourself A Poisson distribution with parameter n has mean and variance both equal to n. Write $\mathrm{P}(X = n)$ exactly and approximately.

Try this next

1. **Check the formula.** Use Stirling's formula to estimate 10! and 20!, and compare with the true values 3 628 800 and 2 432 902 008 176 640 000.
2. **From (iii) to (ii).** Show that Stirling's formula in part (iii) implies the approximation in part (ii).
3. **Digits of** 100!**.** Use $\log_{10} n! \approx n \log_{10} \frac{n}{e} + \frac{1}{2} \log_{10}(2\pi n)$ to show that 100! has 158 digits.
4. **A tie in coin tossing.** Use part (ii) to show that the probability of exactly 500 heads in 1000 tosses of a fair coin is about 0.025.

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter II, which proves Stirling's formula.
- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Chapter 9, on asymptotics.
- **Stephen Stigler, *The History of Statistics*** (Harvard University Press), on de Moivre and the birth of the normal curve.

The full guided solution, and what the question leads to at university, is in Section 12.23: https://beyondstep.keithmcnulty.org/ch12/stirling-probability.html